How-To Library

Informatica

How to Create Cloudera Altus Clusters with a Cluster Workflow in Big Data Management

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Abstract

You can implement Cloudera Altus clusters hosted on Amazon Web Services (AWS) with Informatica® Big Data Management 10.2.1. Create a workflow with a Command task that runs scripts to create the cluster, and Mapping tasks to run mappings on the cluster. You can add another Command task to terminate and delete the cluster when workflow tasks are complete.

Supported Versions

- Informatica® Big Data Management 10.2.1

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Overview

You can use a workflow to create a Cloudera Altus cluster on Amazon Web Services (AWS) and run Mapping and other tasks.

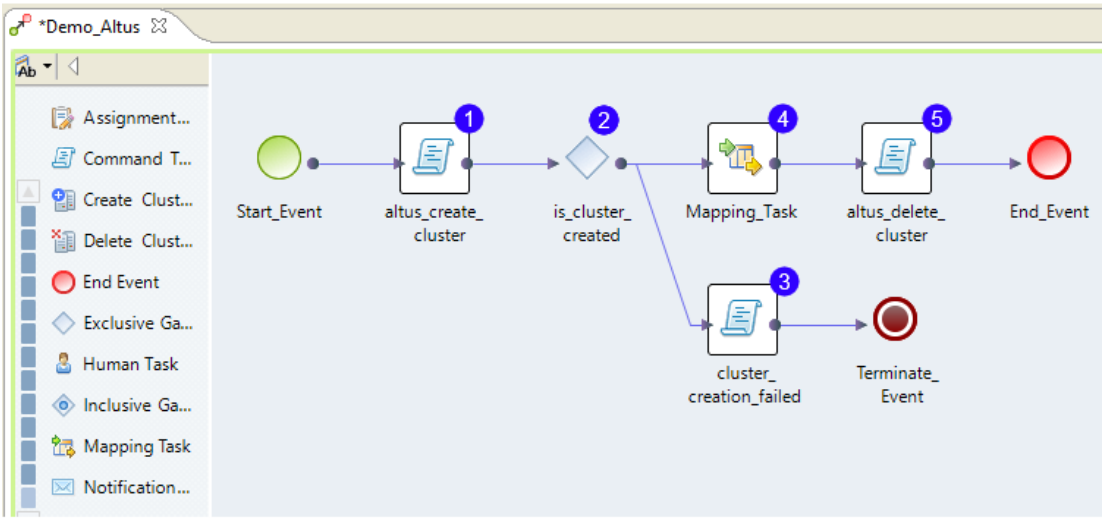
When you include a Command task to terminate and delete the cluster and the resources it used in the cloud platform, the cluster is called an ephemeral cluster. In an ephemeral cluster strategy, cloud clusters are created, exist for the time it takes for jobs to complete, and then cease to exist when they are brought down.

Create a workflow to implement the strategy. The workflow contains Command tasks that use scripts that spin up and configure a Cloudera Altus cluster on AWS, and Mapping tasks that run mappings. When the mapping runs are

complete, an additional Command task can then terminate the cluster, so that you use cluster resources only when you need them.

Cloudera Altus clusters support only mappings that use the Spark run-time engine.

The following image shows a typical workflow to create an Altus cluster, run mappings, and terminate the cluster:



The workflow contains Command and Mapping tasks that run scripts, as well as a gateway event that evaluates the success or failure of tasks. The following descriptions explain elements in the image:

1. The script creates the cluster and configures connections between the cluster and the Informatica domain.

   The Command task contains the altusCreateCluster.sh script. The script uses Cloudera APIs to create an Altus cluster on AWS EC2 instances. The script contains a step to get the IP address of the Cloudera manager node of the created cluster, and uses the address to create a cluster configuration object and a Hadoop connection that enables the Data Integration Service to run jobs on the cluster.
2. An optional gateway event determines whether the previous Command task was successful.
3. If the previous Command task is unsuccessful, you can configure a Command task to handle the event. In this case, the Command task might include a step to delete the cluster and connections.
   For an example of a script to delete the cluster and connections, see "Add a Command Task to Delete the Cluster" on page 12.
4. The workflow runs one or more Mapping tasks, which run mappings on the cluster.
5. You can include a Command task to terminate and delete the cluster. You might want to do this to conserve cloud resource costs.

   The Command task contains the altusDeleteCluster.sh script. The script deletes the Altus cluster, as well as connections and configuration files that are no longer necessary.

   If you do not include a task to terminate and delete the cluster, it continues to run.

Before You Begin

Before you can prepare the AWS environment and create the cluster workflow, you must perform the following tasks:

1. Download the script archive file.
2. Verify prerequisites.

Download the Script Archive File

Download the script archive file Altus_Script_Files.zip. The archive contains script and properties files to edit with property values for the cluster to create, and other files to enable script files to run.

The Altus_Script_Files.zip archive contains the following files:

Properties files to edit:

- arguments.properties
- tags.list

Script files:

- altusClusterCreate.sh
- altusClusterDelete.sh

Python files that the altusClusterCreate.sh and altusClusterDelete.sh files use:

- infatag.py
- pyUtils.py

Note: Do not edit the Python or script files. Simply upload them with the edited files in "Upload Scripts and Properties Files to the EC2 Instance" on page 8.

Download the script archive from Informatica Network at the following URL:
https://kb.informatica.com/h2l/HowTo%20Library/1/Altus_Script_Files.zip.

Verify Prerequisites

Verify the following prerequisites:

- You have an AWS account with a Virtual Private Cloud (VPC) where you have permissions to create resources.

  Read the Cloudera documentation at the following URL for AWS account requirements and settings:
  https://docs.cloudera.com/cdp/latest/requirements-aws/topics/mc-requirements-aws.html.
- You have installed Big Data Management and the Informatica domain on an Amazon EC2 instance in the VPC.

Prepare the Environment and Configure Properties

Before you create the workflow, perform the following tasks:

1. Get AWS account and environment information.
2. Create the Altus environment on AWS.
3. Install software on the Informatica domain instance.
4. Create properties files.
5. Create parameters and configure mappings.
6. Create scripts for workflow gateway events.
7. Upload scripts and properties files to the EC2 instance.
8. Configure custom properties on the Data Integration Service.

Get AWS Account Information

Get AWS account information to use when you configure the AWS environment.

Get the following information from the AWS administrator:

Property	Description
Access Key ID	The access key ID allows the user to make programmatic calls to AWS from the AWS Command Line Interface (AWS CLI), tools for Windows PowerShell, the AWS SDKs, or direct HTTP calls using the APIs for individual AWS services. For more information, see the AWS documentation.
Secret Access Key	The secret access key for access to Amazon account resources. The secret access key is associated with the access key ID and uniquely identifies the account. Required if you do not use AWS Identity and Access Management (IAM) authentication.
Default region	AWS region where the Altus environment and the Informatica domain will reside. Use AWS region values. For a list of acceptable values, see the AWS documentation.

Supply the values for these properties when you create the Altus environment and configure AWS for the Altus cluster.

Create the Altus Environment on AWS

Create a Cloudera Altus environment in the same VPC as the Informatica domain.

Create the Altus environment with appropriate permissions and resources. For more information, see the Cloudera documentation.

As part of the setup, perform the following tasks:

- Install and configure the Altus client in the VPC. For more information, see the Cloudera documentation.
- Configure AWS.
  When you run the configureAWS command, supply the following information:
  - Access Key ID
  - Secret Access Key
  - Default region

As part of the setup, you create a directory on the Data Integration Service host you install the Altus client. Note the directory path, and supply it when you configure custom properties on the Data Integration Service.

Install Software on the Informatica Domain Instance

Install the following software utilities and packages on the Amazon EC2 instance that hosts the domain:

- AWS Command Line Interface (CLI) for Linux. For more information, see the AWS documentation.
- AWS Software Development Kit (SDK) for Python (Boto3). For more information, see the AWS documentation.
- PIP python manager utility. For more information, see the AWS documentation.

Create Properties Files

Create properties files based on templates in the Altus_Script_Files.zip archive. The properties files supply values to workflow Command tasks to create and configure the Altus cluster and connections with the Informatica domain.

Create the following files:

arguments.properties

The arguments.properties file contains properties that the Create Cluster script task uses to create the Altus cluster. Supply a value for each property in the Informatica and Altus sections.

tags.list

The tags.list file allows you to optionally tag the EC2 instances of the Altus cluster. When you tag the EC2 resources, you can easily identify them in resource logs and dashboards. Tags must follow naming conventions.

The tags.list file uses the following format for tags:

```
<tag_name>,<tag_value>,[is_tag_unique]
```

The following table describes the elements of the file:

Property	Description
tag_name	Name of the tag.
tag_value	Value for the tag.
is_tag_unique	Boolean. Indicates if the tag is be unique to all of the EC2 instances that you provision for the cluster. Use the following values for TRUE and FALSE: - T - F

For more information, see the AWS documentation.

Create Parameters and Configure Mappings

Create parameters for the cluster name and the Hadoop connection. The Create Cluster task uses these parameters when creating the cluster and configuring the connection between the domain and the cluster.

Create Mapping Parameters

Parameters enable mappings to inherit and reuse values for mapping properties.

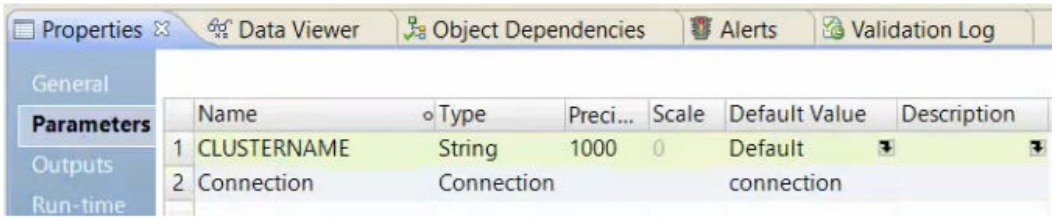
You can open a mapping and create parameters for that mapping and other mappings to use.

To create parameters, complete the following steps:

1. Open a mapping in the Developer tool.
2. In the **Properties** tab, create two new parameters with the following properties:

Parameter Name	Type	Value
CLUSTERNAME	String	Default
Connection	Connection	connection

The following image shows the configured parameters:



Configure Mapping Run-Time Properties

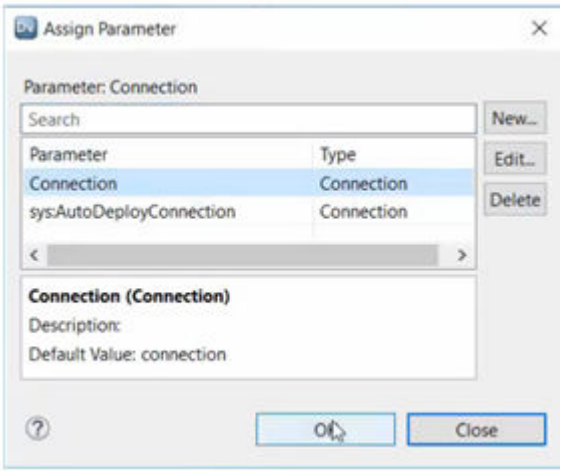
For each mapping that you want to run in the workflow, complete the following steps:

1. In the mapping **Run-Time Properties** tab, set the following run-time engines in the Validation Environment section:
   - Deselect Native.
   - Select Spark.

   Cloudera Altus clusters support only mappings that use the Spark run-time engine.
2. In the Execution Environment section, complete the following steps to assign parameters to the mapping:
   a. Click **Hadoop** > **Connection** and choose **Assign Parameter** from the drop down list.

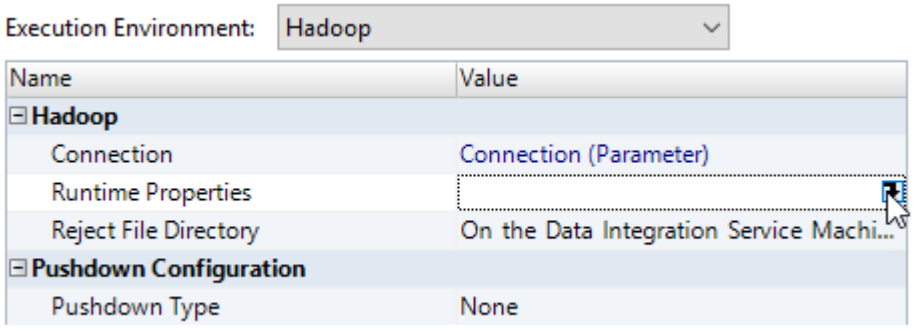
      The **Assign Parameter** dialog box opens.
   b. Select the Connection parameter that you created and click **OK**.

      The following image shows how the Connection parameter appears in the dialog box:



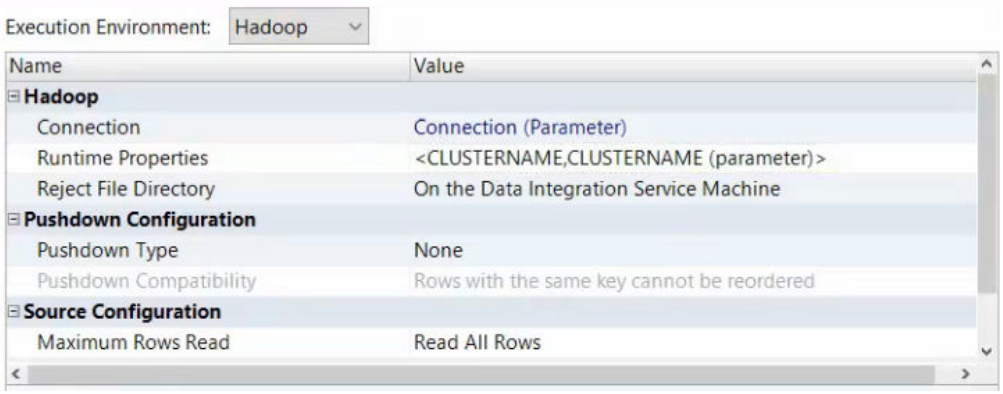
3. Click in the Hadoop Runtime Properties value pane to display the Execution Parameters dialog box.

   The following image shows the selection icon to click:



4. In the **Execution Parameters** dialog box, complete the following steps to create a new parameter:
   a. Click **New**.
   b. Name the new parameter CLUSTERNAME.
   c. Click the selection arrow on the right side of the value pane to display the **Parameters** list.
   d. Click **Assign parameter**.
   e. Select the CLUSTERNAME parameter that you created in step 4b and click **OK**.

   The following image shows the configured run-time mapping properties:



5. Save the mapping.

Create Scripts for Workflow Gateway Events

If you designed the workflow to contain gateway events like the ones in the sample workflow in the overview, create the scripts for the gateway events to execute.

Gateway events allow the workflow to deal with exceptions and other events. You can create scripts to handle events that gateway events trigger, such as the Cluster Creation Failed event that appears in the sample workflow in the Overview.

For more information about gateway events and how to configure them, see the *Developer Workflow Guide.*

Upload Scripts and Properties Files to the EC2 Instance

The cluster workflow uses scripts and properties files to create and configure the Altus cluster and its resources.

Upload the following files to the EC2 instance that hosts the Informatica domain:

- The Create Cluster script altusClusterCreate.sh.

- Scripts to handle gateway events, if you created gateway events in the workflow.
- The Delete Cluster script altusClusterDelete.sh.
- Python scripts (infatag.py and pyUtils.py) that the Create Cluster and Delete Cluster scripts use.
- The arguments.properties file and the tags.list file that you configured.

Configure Custom Properties on the Data Integration Service

Configure custom Data Integration properties to enable the connection to the Altus cluster.

1. In the Administrator tool **Domain Navigator**, select the Data Integration Service that you want to use to run workflows on the Altus cluster.
2. Click the Edit icon to edit the custom properties for the Data Integration Service.

   The following image shows the Edit icon for Custom Properties:



3. The following table shows the properties to add:

Property	Description
AltusCliEnvDir	Path to the Altus client directory on the Data Integration machine. Example: `home/<user directory>/<Altus client directory>`
AltusExecution	Boolean. Set to `true`.

4. Click **OK** to save the custom properties, and then restart the Data Integration Service.

Create the Workflow

Create a workflow with Command tasks to create an Altus cluster and Mapping tasks to run mappings on the cluster. Optionally add a Command task to delete the cluster.

Complete the following steps:

1. Create the workflow and configure workflow parameters.
2. Add a Command task to create and configure the cluster.
3. Add Mapping tasks to the workflow.
4. Optional: Add gateway events in the workflow.
5. Optionally add a Command task to delete the cluster.

Create the Workflow and Configure Workflow Parameters

Create the workflow and create parameters for workflow tasks to use.

When you create workflow parameters, any Command task and its script can use these parameters.

1. Create a workflow in the Developer tool.
2. Click on the Parameters tab of the Workflow properties and create the following parameters:

clusterName

Name of the Altus cluster to create.

nodeCount

Number of EC2 worker nodes in the cluster. Minimum is 3.

Note: By default, in addition to the worker nodes specified in this parameter, the Altus API creates a cluster manager node and a master node. For example, if you specify 3 for the nodeCount, the cluster contains a total of 5 EC2 nodes.

instanceType

EC2 instance type for cluster nodes. Recommended type: m4.xlarge.

For more information about the EC2 node types you can use, see Altus documentation.

cdhVersion

Cloudera version to use with the cluster. Specify CDH511 or later.

For a full list of supported versions, see the Product Availability Matrix on the Informatica network.

altusEnvironment

Name of the Altus environment that you created.

serviceType

Run-time engine types that can run mappings. Set the value to MULTI to enable the Data Integration Service to run jobs using required run-time engines.

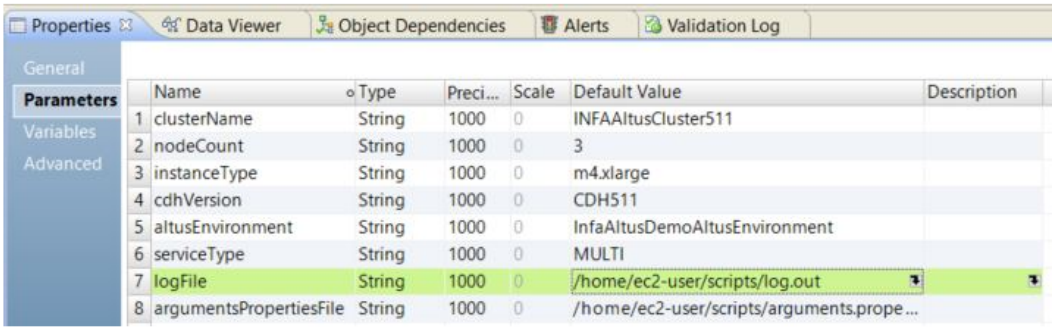
logFile

Location of the log file to create for the cluster creation and deletion process.

argumentsPropertiesFile

Location of the arguments.properties file that you uploaded.

The following image shows sample configured parameters:



Add a Command Task to Create the Cluster

Create a Command task to create and configure the Altus cluster.

Populate the Command task with the following command to enable the task to run the altusClusterCreate.sh script:

```
sh /<path>/altusClusterCreate.sh -c ${par:clusterName} -n ${par:nodeCount} -i $
{par:instanceType} -v ${par:cdhVersion} -e ${par:altusEnvironment} -s ${par:serviceType} -f $
{par:argumentsPropertiesFile} -o ${par:logFile}
```

The script creates the cluster and configures the cluster using inputs from both the altusClusterCreate.sh script and the arguments.properties file.

Add Mapping Tasks in the Workflow

The workflow uses Mapping tasks to run mappings.

You can use mappings that you prepared previously in the step ""Create Parameters and Configure Mappings" on page 6."

1. Drag a Mapping task from the task list to the workflow editor.

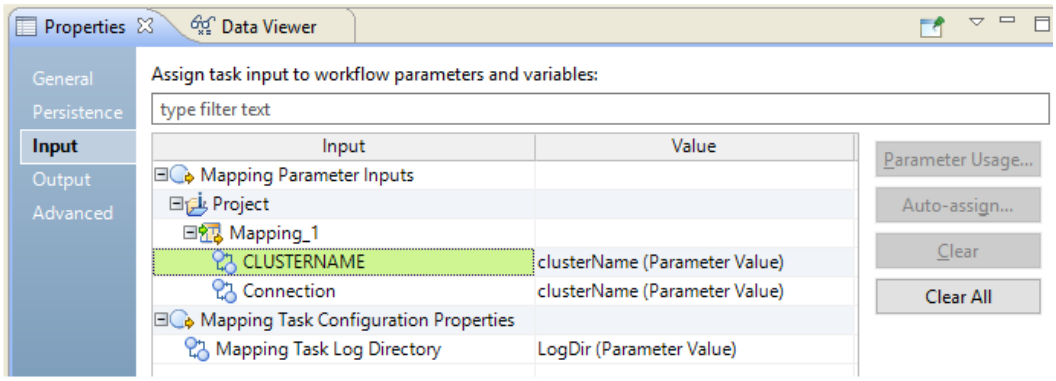
   The Mapping Task dialog box opens.
2. Name the Mapping task.
3. Select one of the mappings that you configured to run with the Mapping task.

   Click **Browse** next to the Mapping property, select a mapping, and click **Finish**.
4. Select the **Input** properties tab and assign the following properties:

Input	Value
CLUSTERNAME	clusterName (Parameter Value)
Connection	clusterName (Parameter Value)

5. Save the Mapping task.

The following image shows the configured Input properties:



Create Gateway Events in the Workflow

Gateway events enable the workflow to detect and handle process failures. When a failure occurs, the workflow branches to one of the Command tasks that you created to handle the failure.

If you designed the workflow to contain gateway events, add the gateway events to the workflow and configure them to run the scripts that you uploaded.

Add a Command Task to Delete the Cluster

Create a Command task if you want to delete the cluster at the end of the workflow.

Populate the Command task with the following command to enable the task to run the altusClusterDelete.sh script:

```
sh /<path>/altusClusterDelete.sh -c ${par:clusterName} -f ${par:argumentsPropertiesFile} -o ${par:logFile}
```

The script deletes the Hadoop connection and the cluster configuration, then deletes the cluster.

Run the Workflow and Monitor Workflow Logs

Deploy the workflow to the Data Integration Service, which runs it automatically. The workflow runs the tasks that you configured.

Deploy a workflow from the Developer tool to the Data Integration Service. To deploy a workflow, contain it in an application. For more information about application deployment, see the *Developer Tool Guide*.

Use the Administrator tool to monitor workflow tasks. You can also monitor jobs on the Cloudera Altus web console.

Monitor Command task execution in the logs at the location that you specified in the Create Cluster command task. Monitor Mapping task execution in the Mapping task log.

For more information about monitoring workflows, see the *Developer Workflow Guide*.

Author

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